Building systems

We're working toward learning to solve systems of differential equations, but before we can, we need to define the kind of system we're trying to solve.

$$x_1'(t) = a_{11}(t)x_1(t) + a_{12}(t)x_2(t) + \ldots + a_{1n}(t)x_n(t) + f_1(t)$$

$$x_2'(t) = a_{21}(t)x_1(t) + a_{22}(t)x_2(t) + \ldots + a_{2n}(t)x_n(t) + f_2(t)$$

...

$$x_n'(t) = a_{n1}(t)x_1(t) + a_{n2}(t)x_2(t) + \ldots + a_{nn}(t)x_n(t) + f_n(t)$$

This is a system of first order linear differential equations. When all of the values $f_i(t)$ are zero, the system is a **homogeneous system**. But if any of the values $f_i(t)$ are non-zero, then the system is a **nonhomogeneous system**.

Converting to matrix form

We already know from earlier how to convert simple systems into matrix form. For example, converting the system

$$3x_1 - 4x_2 = 2$$

$$x_1 + 5x_2 = -1$$

into the matrix form

$$\begin{bmatrix} 3 & -4 \\ 1 & 5 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 2 \\ -1 \end{bmatrix}$$

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is done by creating the coefficient matrix A, multiplying it by a vector of the variables $\vec{x}$, and equating that to a vector of the values from the right side. In other words, we could generalize the matrix equation as

$$A\vec{x} = \vec{b}$$

If we extend this same idea to systems of differential equations, then for the system of differential equations that we defined earlier,

$$x_1'(t) = a_{11}(t)x_1(t) + a_{12}(t)x_2(t) + \ldots + a_{1n}(t)x_n(t) + f_1(t)$$

$$x_2'(t) = a_{21}(t)x_1(t) + a_{22}(t)x_2(t) + \ldots + a_{2n}(t)x_n(t) + f_2(t)$$

$$\ldots$$

$$x_n'(t) = a_{n1}(t)x_1(t) + a_{n2}(t)x_2(t) + \ldots + a_{nn}(t)x_n(t) + f_n(t)$$

we can define

$$\vec{x} = \begin{bmatrix} x_1(t) \\ x_2(t) \\ \vdots \\ x_n(t) \end{bmatrix} \qquad A = \begin{bmatrix} a_{11}(t) & a_{12}(t) & \cdots & a_{1n}(t) \\ a_{21}(t) & a_{22}(t) & \cdots & a_{2n}(t) \\ \vdots & & & \vdots \\ a_{n1}(t) & a_{n2}(t) & \cdots & a_{nn}(t) \end{bmatrix} \qquad F = \begin{bmatrix} f_1(t) \\ f_2(t) \\ \vdots \\ f_n(t) \end{bmatrix}$$

With these values, we can write the system of differential equations as

$$\vec{x}' = A\vec{x} + F$$

And in this form, it's even easier to distinguish homogeneous and nonhomogeneous systems. The system will be homogeneous when F is the zero vector $F = \vec{O}$, and nonhomogeneous when F is non-zero.

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Therefore, a homogeneous system of differential equations can be written in matrix form as

$$\vec{x}' = A\vec{x}$$

Let's do an example where we convert the system of differential equations into matrix form.

Example

Rewrite the system of differential equations in matrix form.

$$x_1' = 3x_1 - 2x_2$$

$$x_2' = -x_1 + 4x_2$$

To write the system in matrix form $\vec{x}' = A\vec{x}$, we define the vector $\vec{x}'$, matrix A, and vector $\vec{x}$ as

$$\vec{x}' = \begin{bmatrix} x_1' \\ x_2' \end{bmatrix} \qquad A = \begin{bmatrix} 3 & -2 \\ -1 & 4 \end{bmatrix} \qquad \vec{x} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

Then we can write the system in matrix form.

$$\vec{x}' = A\vec{x}$$

$$\vec{x}' = \begin{bmatrix} 3 & -2 \\ -1 & 4 \end{bmatrix} \vec{x}$$

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Converting linear differential equations into systems

Higher order linear differential equations, both homogeneous and nonhomogeneous, can be converted into systems of differential equations.

The order of the differential equation will determine the number of equations in the system. In other words, we'll convert second order equations into a system of two equations, a third order equation will become a system of three equations, etc.

In general, we'll start by substituting $x_1(t) = y(t)$, $x_2(t) = y'(t)$, $x_3(t) = y''(t)$, and so on. And we'll always use the differential equation itself to substitute for the highest degree derivative.

If we're presented with initial conditions as part of our problem, we can convert those as well. Let's look at an example so that we can walk through this process step-by-step.

Example

Convert the third order linear differential equation into a system of differential equations.

$$y''' - 3y'' + y' - 2y = 3t^2$$

Our standard procedure here will be to start by defining

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$$x_1(t) = y(t)$$

$$x_2(t) = y'(t)$$

$$x_3(t) = y''(t)$$

Now we need to solve the original differential equation for $y'''(t)$.

$$y''' - 3y'' + y' - 2y = 3t^2$$

$$y''' = 3y'' - y' + 2y + 3t^2$$

Then if we take the derivatives of the equations for $x_1(t)$, $x_2(t)$, and $x_3(t)$, we get

$$x_1'(t) = y'(t) = x_2(t)$$

$$x_2'(t) = y''(t) = x_3(t)$$

$$x_3'(t) = y'''(t) = 3y'' - y' + 2y + 3t^2 = 3x_3(t) - x_2(t) + 2x_1(t) + 3t^2$$

Simplifying these equations gives us a system of equations that's equivalent to the original third order differential equation.

$$x_1' = x_2$$

$$x_2' = x_3$$

$$x_3' = 3x_3 - x_2 + 2x_1 + 3t^2$$

And if we wanted to write this nonhomogeneous system as a matrix equation, we would get

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$$\vec{x}' = A\vec{x} + F$$

$$\vec{x}' = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 2 & -1 & 3 \end{bmatrix} \vec{x} + \begin{bmatrix} 0 \\ 0 \\ 3t^2 \end{bmatrix}$$

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